

Fatemeh Doudi

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Education

Ph.D. in Computer Engineering

Texas A&M University

Jan 2023 – Present

Advisor: Prof. Dileep Kalathil

GPA: 4.0/4.0

M.Sc. in Electrical Engineering

Sharif University of Technology

Oct 2020 – Dec 2022

GPA: 17.78/20

B.Sc. in Electrical Engineering

Sharif University of Technology

Sep 2015 – Dec 2019

GPA: 16.77/20

Research Experience

Hierarchical Reinforcement Learning for Multi-Agent Orchestration

Texas A&M University

Aug. 2026 – Present

Ongoing research

- Developing a hierarchical RL framework with decomposed high and low level actions.
- Implementing SMDP-based credit assignment.

AgentDiscover: Autonomous Discovery with Minimal Search Scaffolding

Texas A&M University

May 2026 - Sep. 2026

Submitted to ICLR 2027

- Developed a coding-agent framework that plans evolutionary search and proposes candidate programs, using a graph database to retrieve prior ideas and results.
- At matched model spend, outperformed the compared discovery frameworks on a kernel optimization task and produced first-place-equivalent solutions for seven past AtCoder contests.

Reinforcement Learning for Diffusion LLMs with Entropy-Guided Step Selection and Stepwise Advantages

Texas A&M University

Aug. 2025 – June 2026

Accepted to NeurIPS 2026

- Formulated diffusion LLM generation as a finite-horizon MDP over denoising steps and derived an exact policy gradient with stepwise advantages.
- Proposed Entropy-Guided Stepwise Policy Optimization (EGSPO) for compute-efficient RL fine-tuning of diffusion LMs.
- Achieved consistent improvements on coding and reasoning benchmarks (MBPP, HumanEval, Sudoku, GSM8K, MATH500).

Inference-Time Multi-Preference Alignment for Diffusion Models

Texas A&M University

Dec. 2024 – May 2025

Supervisors: Prof. Dileep Kalathil, Prof. P. R. Kumar

Accepted to ICLR 2026

- Proposed Diffusion Blend, an inference-time alignment framework enabling multi-objective reward blending without retraining.
- Achieved fine-tuned-level alignment performance while reducing alignment compute cost by 40%.
- Implemented and benchmarked gradient guidance, inference search, and KL-controlled reward blending methods.

Time Series Prediction in Electric Power Systems using Deep State Space Model

Texas A&M University

July. 2024 – Dec. 2024

Supervisors: Prof. Dileep Kalathil, Prof. Le Xie

Published in *IEEE Transactions on Power Systems*

- Designed PowerMamba, a deep state-space model for electric grid time-series forecasting.
- Achieved 7% accuracy improvement with 43% fewer parameters compared to transformer baselines.
- Implemented large-scale training and evaluation pipelines for spatio-temporal forecasting.

Skills

- **Programming Languages:** Python (primary), C/C++, MATLAB.
- **Agentic AI:** Coding agents, tool use, MCP servers, multi-agent orchestration, hierarchical planning, agent memory, evolutionary search.
- **Machine Learning & LLMs:** Diffusion language models, autoregressive LLMs, reinforcement learning, RLHF, inference-time alignment, reasoning, preference optimization, retrieval-augmented generation (RAG).
- **Modeling & Architectures:** State-space models (Mamba), transformers, diffusion models, bandits, stochastic processes.
- **Systems & Tooling:** PyTorch, distributed training, large-scale experimentation, model evaluation and benchmarking, experiment tracking.
- **Theory:** Reinforcement learning, stochastic systems, game theory, convex and non-convex optimization.

Publications

Reinforcement Learning for Diffusion LLMs with Entropy-Guided Step Selection and Step-wise Advantages

V. T. Kunde*, F. Doudi*, M. Farahbakhsh, D. Kalathil, K. Narayanan, J.-F. Chamberland

NeurIPS 2026 (accepted)

arXiv:2603.12554

(*equal contribution)

Diffusion Blend: Inference-Time Multi-Preference Alignment for Diffusion Models

M. Cheng, F. Doudi, D. Kalathil, M. Ghavamzadeh, P. R. Kumar

ICLR 2026 (accepted)

arXiv:2505.18547

PowerMamba: A Deep State Space Model and Comprehensive Benchmark for Time Series Prediction in Electric Power Systems

A. Menati, F. Doudi, D. Kalathil, L. Xie

IEEE Transactions on Power Systems, 2025

Exploring the Capabilities and Limitations of Large Language Models in the Electric Energy Sector

S. Majumder*, L. Dong*, F. Doudi*, Y. Cai*, C. Tian, D. Kalathil, K. Ding, A. A. Thatte

Joule, 2024.

(*equal contribution)

Relevant Courses

- **Machine Learning & AI:** ML, Deep Learning, Reinforcement Learning, Computer Vision
- **Optimization & Algorithms:** Convex Optimization, Advanced Optimization, Analysis of Algorithms
- **Stochastic Systems & Theory:** Stochastic Systems, Game Theory, Queueing Theory, Bandit Algorithms

Achievements and Awards

- **ECEN Merit Fellowship**, Texas A&M University
- Ranked **4th nationwide** in M.Sc. National Entrance Exam (20,000 participants)
- Ranked **57th nationwide** in B.Sc. National Entrance Exam (180,000 participants)